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EXP32:

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#include <stdio.h>
#include <stdlib.h>
#include <string.h>

typedef unsigned long long u64;

u64 modpow(u64 b, u64 e, u64 m) {
    u64 r = 1; b %= m;
    while (e) { if (e & 1) r = r * b % m; b = b * b % m; e >>= 1; }
    return r;
}
long long inv(long long a, long long m) {
    long long x;
    a = ((a % m) + m) % m;
    for (x = 1; x < m; x++) if ((a * x) % m == 1) return x;
    return -1;
}
long long md(long long a, long long m) { return ((a % m) + m) % m; }

/* toy hash of a message */
long long hash(const char *s, long long mod) {
    long long h = 7; int i;
    for (i = 0; s[i]; i++) h = (h * 31 + s[i]) % 100003;
    return h % mod;
}

/* toy DSA parameters: p = 283, q = 47 (q | p-1), g = 64 (order q) */
u64 p = 283, q = 47, g = 64, x = 24, y;
```

```
void dsa_sign(const char *m, int k, long long *r, long long *s) {
    long long h = hash(m, q);
    *r = modpow(g, k, p) % q;
    *s = md(inv(k, q) * (h + x * (*r)), q);
}
int dsa_verify(const char *m, long long r, long long s) {
    long long h = hash(m, q), w = inv(s, q), u1 = md(h * w, q), u2 = md(r * w, q);
    long long v = (modpow(g, u1, p) * modpow(y, u2, p) % p) % q;
    return v == r;
}

int main() {
    const char *m = "Pay Bob 100 rupees";
    long long r1, s1, r2, s2, k, h1, h2, xr;
    long long n = 3233, e = 17, d = 2753;           /* toy RSA key (61*53) */
    int tries = 0;
    srand(7);
    y = modpow(g, x, p);

    printf("== DSA: same message signed twice (fresh random k each time) ==\n");
    do { dsa_sign(m, 1 + rand() % (q - 1), &r1, &s1); } while (r1 == 0 || s1 == 0);
    do { dsa_sign(m, 1 + rand() % (q - 1), &r2, &s2); } while (r2 == 0 || s2 == 0 || (r2 == r1
&& s2 == s1));
    printf("Signature 1 = (r=%lld, s=%lld)  valid: %s\n", r1, s1, dsa_verify(m, r1, s1) ? "yes" :
"no");
    printf("Signature 2 = (r=%lld, s=%lld)  valid: %s\n", r2, s2, dsa_verify(m, r2, s2) ? "yes" :
"no");

    printf("\n== RSA: same message signed twice ==\n");
    h1 = hash(m, n);
```

```
    printf("Signature 1 = %llu\nSignature 2 = %llu   (always identical)\n",
           modpow(h1, d, n), modpow(h1, d, n));

    printf("\n== Implication: k must be secret and never reused in DSA ==\n");
    k = 10;
    dsa_sign("Message A", k, &r1, &s1);
    dsa_sign("Message B", k, &r2, &s2);
    h1 = hash("Message A", q); h2 = hash("Message B", q);
    printf("Two messages signed with the same k give r = %lld and %lld (equal r)\n", r1, r2);
    k = md((h1 - h2) * inv(s1 - s2, q), q);
    xr = md((s1 * k - h1) * inv(r1, q), q);
    printf("Attacker recovers k = %lld and the PRIVATE key x = %lld (true x = %llu)\n", k, xr,
x);
    return 0;
}
```

OUTPUT:

```
== DSA: same message signed twice (fresh random k each time) ==
Signature 1 = (r=42, s=5)  valid: yes
Signature 2 = (r=29, s=29)  valid: yes

== RSA: same message signed twice ==
Signature 1 = 2262
Signature 2 = 2262   (always identical)

== Implication: k must be secret and never reused in DSA ==
Two messages signed with the same k give r = 15 and 15 (equal r)
Attacker recovers k = 10 and the PRIVATE key x = 24 (true x = 24)
```